$\phi : \mathrm{Hom}(X, ?) \to \mathrm{Hom}(Y, ?)$ *there is a unique isomorphism* $a_\phi : X \to Y$ *inducing* ϕ.

Proof (Sketch). One sets $a_\phi = \phi_Y^{-1}(1_Y)$ and shows that it is invertible by constructing the inverse, which is $a_\phi^{-1} = \phi_X(1_X)$. It remains to show that the composition both ways is the identity, which we will omit here. This establishes the existence of a_ϕ. Its uniqueness is verified in a straightforward manner. □

Remark 7.5.2. In a similar way, if a category $\mathcal{C}$ is enriched over another category $\mathcal{D}$ (say, the category of abelian groups or vector spaces), one can define the notion of a representable functor from $\mathcal{C}$ to $\mathcal{D}$.

Example 7.5.3. Let A be an algebra. Then the forgetful functor from the category of left A-modules to the category of vector spaces is representable, and the representing object is the free rank 1 module (= the regular representation) $M = A$. But if A is infinite dimensional and we restrict attention to the category of finite dimensional modules, then the forgetful functor, in general, is not representable (this is so, for example, if A is the algebra of complex functions on $\mathbb{Z}$ which are zero at all but finitely many points).

7.6. Adjoint functors

Another fundamental notion in category theory is the notion of adjoint functors.

Definition 7.6.1. Functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are said to be a pair of **adjoint functors** if for any $X \in \mathcal{C}$, $Y \in \mathcal{D}$ we are given an isomorphism $\xi_{XY} : \mathrm{Hom}_{\mathcal{D}}(F(X), Y) \to \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$ which is functorial in X and Y, i.e., if we are given an isomorphism of functors $\mathrm{Hom}_{\mathcal{D}}(F(?), ?) \to \mathrm{Hom}_{\mathcal{C}}(?, G(?))$ ($\mathcal{C} \times \mathcal{D} \to \mathbf{Sets}$). In this situation, we say that F is **left adjoint** to G and G is **right adjoint** to F.

Not every functor has a left or right adjoint, but if it does, it is unique and can be constructed canonically (i.e., if we somehow found two such functors, then there is a canonical isomorphism between them). This follows easily from the Yoneda lemma, since if